# Esp32Tap — Depth-First Cluster Build and Test

**Guide mode:** Empty-board build · Build one cluster, prove it, sign it, then unlock the next.

No treadmill cable before clusters 1–10 pass; do not connect a treadmill cable until Cluster 11. No exact physical placement is prescribed. Follow the named parts, pins, nets, and colors.

| | |
|---|---|
| **Device serial:** | **Operator:** |
| **Hardware revision:** | **Date:** |

**Source manifest:** esp32tap-breadboard-netlist.csv; esp32tap-breadboard-clusters.html; esp32tap-breadboard-from-to.html; DigiKey invoice 129832848; superseded module checklist used only for already-proven limits and sequences.

## Source contract

**Mutually exclusive source rule:** USB and STANDALONE POWER are mutually exclusive—one source only. USB connected means STANDALONE POWER physically removed. STANDALONE POWER installed means USB physically disconnected. All sources off before any RJ45 attachment/removal, harness change, or direct-path change. COIL POWER stays open through the unloaded test.

## Safety contract

- Initial source: 8.00 V, initial 250 mA current limit; coil-open current below 50 mA. VIN must be 7.20–7.90 V and LOGIC_3V3 must be 3.20–3.40 V.
- TREAD_OK is low below UV, high at 8.00 V, and low above OV. UV rising is 6.25–6.55 V; OV falling is 10.30–10.90 V.
- TPS709 enabled: 4.75–5.25 V; TPS709 disabled: below 0.25 V. Loaded relay current limit no more than 500 mA.
- Coil: 90–110 mA and at least 4.50 V. BC337 VCE no more than 0.30 V. Feedback (1,0) energized and (0,1) bypass; 00 or 11 is a fault.
- Removal of USB logic power, GPIO21 command, TREAD_OK, or VIN must restore NC bypass within 100 ms.
- Five-minute hold: TPS709 and BC337 no more than 45°C and no more than 10°C over ambient.
- Any unexpected reset is a STOP. No treadmill relay transfer or MCU TX is permitted by this guide.

**Firmware:** The current no-control diagnostic cannot pass a relay exercise gate and cannot pass a UART exercise gate. Relay/coil/TX work requires a bounded relay exerciser with exact identity. Standalone/bypass requires an observer identity and manifest proving relay and TX disabled.

Wire palette: BLACK GND · RED +8V_RAW/FUSED_8V · ORANGE VIN · BLUE TSR_3V3/LOGIC_3V3 · VIOLET +5V_RLY/coil · YELLOW control/UART · GREEN feedback/RX · WHITE direct pass-through.

# Cluster 1 — Raw protection

| | |
|---|---|
| **Dependencies:** Empty board and verified bench equipment<br>**Inputs:** +8V_RAW, GND | **Outputs:** VIN, FUSED_8V<br>**Source state:** USB absent; STANDALONE POWER removed; COIL POWER removed; bench 8.00 V, initial 250 mA limit |

## Parts

**F1** RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar. **D1** 1N5822-TP Schottky · 1N5822-TPMSCT-ND · cathode stripe faces VIN. **D2** P6KE12A-TP · P6KE12A-TPMSCT-ND · cathode to VIN. **CIN** 22 µF 25 V · 732-8821-1-ND · negative stripe to GND.

**Operator part record:** F1 · RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar
**Operator wiring record:** F1 · lead 1 · +8V_RAW · RED

## Build — exact pin/net/color wiring

Wire F1 lead 1 to +8V_RAW in RED; F1 lead 2 to FUSED_8V in RED; D1 anode to FUSED_8V and D1 cathode to VIN in ORANGE; D2 cathode and CIN positive to VIN in ORANGE; D2 anode and CIN negative to GND in BLACK.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Check polarity and resistance · polarity · D1 cathode/VIN, D2 cathode/VIN, CIN negative/GND, VIN–GND · stripes correct and no short | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Apply protected raw power · +8V_RAW at 8.00 V · FUSED_8V and VIN · record volts/current · VIN 7.20–7.90 V; coil-open current below 50 mA | |

**STOP gate:** Current approaches 250 mA, VIN is outside 7.20–7.90 V, polarity is wrong, or anything warms. Remove power; do not build Cluster 2.

**PASS gate:** Protection chain, polarity, VIN window, and coil-open current pass. **PASS — continue** to Cluster 2 only.

Operator: ______________ Date: ______________ Signed PASS: ______________

# Cluster 2 — TSR supply

| | |
|---|---|
| **Dependencies:** Cluster 1<br>**Inputs:** VIN, GND | **Outputs:** TSR_3V3<br>**Source state:** Cluster 1 source state retained; STANDALONE POWER remains removed |

## Parts

**U2** TSR 1-2433E · 3.3 V converter · 1951-TSR1-2433E-ND · pin order verified. **C2/C3** 10 µF 50 V electrolytics · 1189-2322-ND · negative stripes to GND. **JP1** removable STANDALONE POWER · NOT ORDERED · leave removed.

**Operator part record:** U2/C2/C3/JP1 · TSR 1-2433E; two electrolytics; STANDALONE POWER · 3.3 V; 10 µF 50 V; removable link · 1951-TSR1-2433E-ND; 1189-2322-ND; NOT ORDERED · U2 pin order verified; capacitor negatives to GND; link removed
**Operator wiring record:** U2/C2/C3/JP1 · U2 pins 1/2/3; capacitor positive/negative; removable terminals · VIN/GND/TSR_3V3/LOGIC_3V3 · ORANGE/BLACK/BLUE

## Build — exact pin/net/color wiring

Wire U2 pin 1 and C2 positive to VIN in ORANGE; U2 pin 2 plus both capacitor negatives to GND in BLACK; U2 pin 3 and C3 positive to TSR_3V3 in BLUE. Fit JP1 between TSR_3V3 and the future LOGIC_3V3 endpoint, but leave it removed.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Verify converter map and capacitor polarity · continuity · U2 pins 1/2/3 to VIN/GND/TSR_3V3 · exact mapping; TSR_3V3 not joined to future LOGIC_3V3 | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Energize verified VIN · VIN · TSR_3V3 · record input/output/current · TSR_3V3 3.20–3.40 V and future LOGIC_3V3 unpowered | |

**STOP gate:** TSR_3V3 is outside 3.20–3.40 V, polarity is wrong, LOGIC_3V3 is energized across the removed link, or current/temperature rises.

**PASS gate:** Verified Cluster 1 VIN produces isolated TSR_3V3 in range. **PASS — continue** to Cluster 3 only.

Operator: ______________________ Date: ______________________ Signed PASS: ______________________

# Cluster 3 — DevKit and logic supply

**Dependencies:** Cluster 2; Verified USB bench source
**Inputs:** TSR_3V3, USB_5V, GND

**Outputs:** LOGIC_3V3, DevKit GND
**Source state:** First USB-only with STANDALONE POWER removed; then USB physically disconnected before standalone link installation

## Parts

**U1** ESP32-S3-DEVKITC-1-N8R8 · 1965-ESP32-S3-DEVKITC-1-N8R8-ND · antenna clear, USB ports identified. **JP1** STANDALONE POWER · NOT ORDERED · removable and labeled. **C4** 100 nF · BC1084CT-ND · non-polar.

**Operator part record:** U1/JP1/C4 · ESP32-S3-DEVKITC-1-N8R8; STANDALONE POWER; capacitor · DevKit; removable link; 100 nF · 1965-ESP32-S3-DEVKITC-1-N8R8-ND; NOT ORDERED; BC1084CT-ND · antenna clear; USB ports identified; link removable; capacitor non-polar
**Operator wiring record:** U1/JP1/C4 · DevKit 3V3/GND; JP1 terminals; capacitor leads · LOGIC_3V3/GND/TSR_3V3 · BLUE/BLACK

## Build — exact pin/net/color wiring

Wire DevKit 3V3 to LOGIC_3V3 in BLUE and DevKit GND to GND in BLACK; C4 across LOGIC_3V3/GND. Finish JP1 between TSR_3V3 and LOGIC_3V3 in BLUE. Construct the command links exactly: DevKit GPIO15 ↔ removable GPIO15 jumper ↔ TX_ENABLE / AHC08 pin 4; DevKit GPIO21 ↔ removable GPIO21 jumper ↔ RELAY_CMD / AHC08 pin 1.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Map source handoff · mapping · DevKit 3V3, LOGIC_3V3, TSR_3V3, JP1 · USB path mapped; JP1 removable; no source backfeed | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Prove each exclusive source · USB_5V then TSR_3V3 · LOGIC_3V3 · record both runs · LOGIC_3V3 3.20–3.40 V; never energize USB and JP1 together | |

**STOP gate:** Sources overlap, either source backfeeds the other, LOGIC_3V3 is outside 3.20–3.40 V, or DevKit boot is unstable.

**PASS gate:** USB-only and standalone-only handoff both pass, with the other source physically absent. **PASS — continue** to Cluster 4 only.

Operator: ____________ Date: ____________ Signed PASS: ____________

# Cluster 4 — TPS3700 voltage monitor

| | |
|---|---|
| **Dependencies:** Cluster 1, Cluster 3<br>**Inputs:** VIN, LOGIC_3V3, GND | **Outputs:** UV_SENSE, OV_SENSE, TREAD_OK, TREAD_OK_MCU<br>**Source state:** USB logic only; STANDALONE POWER removed; bench source swept at protected VIN |

## Parts

**U3** TPS3700DDCR · 296-30395-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **RUV** 150 kΩ · 150KXBK-ND; **ROV** 255 kΩ · 13-MFR-25FBF52-255K-ND; 10 kΩ/100 kΩ/4.7 kΩ · NOT ORDERED; two 1 nF · 445-173473-1-ND; 100 nF · BC1084CT-ND.

**Adapter gate:** continuity-map adapter pins 1–6 to the header before installation. Record the mapped header positions: 1 ____ 2 ____ 3 ____ 4 ____ 5 ____ 6 ____.

**Divider formulas:** UV_SENSE = VIN × 10 kΩ / (150 kΩ + 10 kΩ) = VIN / 16. OV_SENSE = VIN × 10 kΩ / (255 kΩ + 10 kΩ) = VIN / 26.5.

**Operator part record:** U3/dividers/filters · TPS3700DDCR plus adapter and UV/OV networks · 150 kΩ/10 kΩ; 255 kΩ/10 kΩ; 1 nF; 100 nF · 296-30395-1-ND; A879AR-ND; 150KXBK-ND; 13-MFR-25FBF52-255K-ND; 445-173473-1-ND; BC1084CT-ND; NOT ORDERED · meter-map adapter pin 1; capacitors non-polar
**Operator wiring record:** U3/dividers/filters · U3 pins 1–6 and resistor/capacitor endpoints · VIN/GND/UV_SENSE/OV_SENSE/TREAD_OK/TREAD_OK_MCU · ORANGE/BLACK/WHITE/YELLOW/BLUE

**Operator header record:** HDR3 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

## Build — exact pin/net/color wiring

ORANGE: VIN to U3 pin 5, 150 kΩ top, 255 kΩ top, bypass. BLACK: U3 pin 2 and divider/filter returns. WHITE: UV_SENSE to pin 3 from 150 kΩ/10 kΩ/1 nF; OV_SENSE to pin 4 from 255 kΩ/10 kΩ/1 nF. YELLOW: pins 1 and 6 tied as TREAD_OK; 10 kΩ pull-up to LOGIC_3V3, 100 kΩ pull-down to GND, 4.7 kΩ to GPIO6 as TREAD_OK_MCU.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Meter divider and adapter · resistance · VIN–UV_SENSE–GND, VIN–OV_SENSE–GND, U3 pins · 150 kΩ/10 kΩ and 255 kΩ/10 kΩ networks; pin map exact | |
| Continuity-map adapter header · continuity · adapter pins 1–6 to header · one-to-one recorded mapping before installation | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Sweep protected VIN · VIN · TREAD_OK · record low/high transitions · low below UV, high at 8.00 V, low above OV; UV rising 6.25–6.55 V; OV falling 10.30–10.90 V | |

**STOP gate:** TREAD_OK three-state behavior or thresholds fail; never exceed protected VIN about 11.1 V; stop if TVS draws appreciable current or warms.

**PASS gate:** Both sweep directions and GPIO6 isolated observation match limits. **PASS — continue** to Cluster 5 only.

Operator: ____________ Date: ____________ Signed PASS: ____________

## Cluster 5 — AHC08 permission logic

| | |
|---|---|
| **Dependencies:** Cluster 3, Cluster 4<br>**Inputs:** LOGIC_3V3, TREAD_OK, RELAY_CMD, TX_ENABLE, GND | **Outputs:** RELAY_GATE, TX_GATE<br>**Source state:** USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection |

**Source state record:** USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

### Parts

**U4** SN74AHC08N PDIP-14 · 296-4524-5-ND · notch verified, across center channel. **C5** 100 nF · BC1084CT-ND · non-polar. **R1/R2** 10 kΩ command pull-downs · NOT ORDERED.

**Operator part record:** U4/C5/R1/R2 · SN74AHC08N; capacitor; pull-downs · PDIP-14; 100 nF; two 10 kΩ · 296-4524-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar
**Operator wiring record:** U4/C5/R1/R2 · U4 pins 1–14 and component endpoints · LOGIC_3V3/GND/RELAY_CMD/TREAD_OK/RELAY_GATE/TX_ENABLE/TX_GATE · BLUE/BLACK/YELLOW

### Build — exact pin/net/color wiring

BLUE U4 pin 14 to LOGIC_3V3; BLACK pin 7 and unused inputs 9,10,12,13 to GND; pins 8/11 no connection. YELLOW: GPIO21/RELAY_CMD to pin 1, TREAD_OK pin 2, RELAY_GATE pin 3; GPIO15/TX_ENABLE pin 4, TREAD_OK pin 5, TX_GATE pin 6. Add 10 kΩ pull-downs and C5 pin 14-to-7.

### Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Check DIP orientation and control mapping · mapping · U4 pins 1–14 and named nets · exact map, unused inputs grounded, outputs not shorted | |

### Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Exercise both AND gates · RELAY_CMD/TX_ENABLE plus TREAD_OK · RELAY_GATE/TX_GATE · record 00,10,01,11 truth tables · output high only for command=1 and TREAD_OK=1 | |

Remove GPIO15 jumper before manual injection of TX_ENABLE. Remove GPIO21 jumper before manual injection of RELAY_CMD. Never drive against a connected DevKit GPIO.

**STOP gate:** Any gate rises without both inputs, a command GPIO remains connected during manual drive, or supply/reset behavior is abnormal.

**PASS gate:** Both four-state truth tables pass and temporary drives are removed. **PASS — continue** to Cluster 6 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

# Cluster 6 — TPS709 and BC337 driver

| | |
|---|---|
| **Dependencies:** Cluster 1, Cluster 5<br>**Inputs:** VIN, RELAY_GATE, GND | **Outputs:** +5V_RLY, RELAY_COIL−<br>**Source state:** Valid VIN/TREAD_OK; COIL POWER remains removed for the unloaded test |

## Parts

**U5** TPS70950DBVR · 296-35484-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **Q1** BC337-40 · 4878-BC337-40CT-ND; **RBASE** 560 Ω · MFR-25FBF52-560R-ND; 10 kΩ · NOT ORDERED; 1 µF · 445-173583-1-ND; 4.7 µF · 732-8660-1-ND; **JP2** COIL POWER · NOT ORDERED.

**Operator part record:** U5/Q1/RBASE/caps/JP2 · TPS70950DBVR; BC337-40; 560 Ω/10 kΩ; 1 µF/4.7 µF; COIL POWER · 5 V regulator and unloaded driver · 296-35484-1-ND; A879AR-ND; 4878-BC337-40CT-ND; MFR-25FBF52-560R-ND; 445-173583-1-ND; 732-8660-1-ND; NOT ORDERED · adapter and Q1 leads mapped; electrolytic negative to GND; link removed
**Operator wiring record:** U5/Q1/RBASE/caps/JP2 · U5 pins 1–5; Q1 B/C/E; component endpoints · VIN/GND/RELAY_GATE/+5V_RLY/RELAY_COIL− · ORANGE/BLACK/YELLOW/VIOLET

**Operator header record:** HDR5 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

## Build — exact pin/net/color wiring

ORANGE VIN to U5 pin 1 and input capacitor; BLACK pin 2/cap returns/Q1 emitter; YELLOW RELAY_GATE to U5 pin 3 and through 560 Ω to Q1 base with 10 kΩ base pull-down; U5 pin 4 no connection; VIOLET pin 5/output capacitor to +5V_RLY, Q1 collector to RELAY_COIL−. Fit labeled JP2 but leave removed.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Verify adapter, transistor leads, and isolation · polarity · U5 pins, Q1 B/C/E, output capacitor, JP2 · exact map; COIL POWER open | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Toggle gate unloaded · RELAY_GATE · +5V_RLY and Q1 base · record enabled/disabled/discharge · TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V | |

**STOP gate:** COIL POWER is installed, enabled or disabled limits fail, output does not discharge below 0.25 V, or Q1/U5 heats.

**PASS gate:** Unloaded regulator and base drive pass with coil isolated. **PASS — continue** to Cluster 7 only.

| Operator: | Date: | Signed PASS: |
|---|---|---|

# Cluster 7 — Relay coil, local contacts, and feedback

**Dependencies:** Cluster 6, Cluster 3
**Inputs:** +5V_RLY, RELAY_COIL−, LOGIC_3V3, GND

**Outputs:** LOCAL_NC, LOCAL_NO, FB_NC, FB_NO
**Source state:** Local-only bench exercise; treadmill/RJ45 cables absent; bounded relay exerciser verified; loaded relay limit no more than 500 mA

## Parts

**K1** G5V-2 DC5 DPDT · 39-G5V-2DC5-ND · bottom-view drawing, meter-map physical pins. **D3** P6KE6.8CA bidirectional · 18-P6KE6.8CACT-ND. **JP2** COIL POWER · NOT ORDERED. **RFB** two 10 kΩ · NOT ORDERED.

**Operator part record:** K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 kΩ · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional
**Operator wiring record:** K1/D3/JP2/RFB · K1 pins 1/16, 4/6/8, 9/11/13 and component endpoints · +5V_RLY/RELAY_COIL−/LOCAL_NC/LOCAL_NO/GND/FB_NC/FB_NO · VIOLET/WHITE/BLACK/GREEN/BLUE

## Build — exact pin/net/color wiring

VIOLET JP2 from +5V_RLY to K1 pin 1, Q1 collector/RELAY_COIL− to pin 16, D3 across pins 1/16. WHITE local contact fixture: NC pin 4, COM pin 6, NO pin 8. BLACK feedback COM pin 11 to GND; GREEN pin 13 to GPIO4/FB_NC and pin 9 to GPIO5/FB_NO, each with BLUE 10 kΩ pull-up to LOGIC_3V3.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Map relay before attaching local fixture · continuity · pins 1–16, 4–6, 8–6, 13–11, 9–11 · coil about 50 Ω; NC pairs closed; NO pairs open | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Exercise bounded relay and fail sources · +5V_RLY/RELAY_COIL− · local contacts/feedback · record current, voltage, VCE, timing, temperature · coil 90–110 mA, voltage at least 4.50 V, BC337 VCE no more than 0.30 V; (1,0) energized, (0,1) bypass; 00 or 11 fault | |

**Bound operator part record:** K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 kΩ · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

**Relay exerciser identity:** ____________________ **Firmware identity:** ____________________

From energized state, removal of USB logic power must release to NC bypass; removal of GPIO21 command must release to NC bypass; removal of TREAD_OK must release to NC bypass; removal of VIN must release to NC bypass. Each NC bypass return must occur in no more than 100 ms. Then hold five minutes: TPS709 and BC337 at or below 45°C and no more than 10°C over ambient.

## Dedicated fail-release and thermal evidence

| Evidence field | Measured | Limit |
|---|---|---|
| USB logic release (ms) | | NC bypass ≤100 ms |
| GPIO21 release (ms) | | NC bypass ≤100 ms |
| TREAD_OK release (ms) | | NC bypass ≤100 ms |
| VIN release (ms) | | NC bypass ≤100 ms |
| Ambient temperature (°C) | | Record before hold |
| TPS709 temperature (°C) | | ≤45°C and ≤10°C over ambient |
| BC337 temperature (°C) | | ≤45°C and ≤10°C over ambient |

**STOP gate:** Any electrical, feedback, release, reset, or five-minute thermal gate fails. Cluster 7 is local-only; it makes no end-to-end RJ45 claim.

**PASS gate:** Local contacts, feedback, current, voltage, release, and thermal evidence all pass. **PASS — continue** to Cluster 8 only.

Operator: ____________ Date: ____________ Signed PASS: ____________

# Cluster 8 — AHC126 and UART taps

| | |
|---|---|
| **Dependencies:** Cluster 3, Cluster 5<br>**Inputs:** LOGIC_3V3, TX_GATE, GPIO17_TX, GND | **Outputs:** TX_DRV, UART_BUFFER_HIGH_Z, UART_BUFFER_FOLLOWS<br>**Source state:** Local bench fixture only; treadmill and both RJ45 breakouts absent; bounded UART exerciser identity recorded |

## Parts

**U6** SN74AHC126N PDIP-14 · 296-4535-5-ND · notch verified. **C6** 100 nF · BC1084CT-ND. **R3** 100 Ω and **R4/R5** 10 kΩ · NOT ORDERED.

**Operator part record:** U6/C6/R3/R4/R5 · SN74AHC126N; capacitor; series/tap resistors · PDIP-14; 100 nF; 100 Ω; two 10 kΩ · 296-4535-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar
**Operator wiring record:** U6/C6/R3/R4/R5 · U6 pins 1–14 and resistor endpoints · LOGIC_3V3/GND/TX_GATE/GPIO17_TX/TX_DRV · BLUE/BLACK/YELLOW

## Build — exact pin/net/color wiring

BLUE U6 pin 14 to LOGIC_3V3; BLACK pin 7 and unused pins 4,5,9,10,12,13 to GND; pins 6,8,11 no connection. YELLOW TX_GATE to pin 1, GPIO17_TX to pin 2, and pin 3 through 100 Ω to the local TX_DRV test point. Do not attach K1, CONSOLE.6, PIN3, GPIO16, or GPIO18 here; connector-side taps and relay-selected TX are deferred to Cluster 11.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Check local buffer map · continuity · U6 pins, 100 Ω resistor, local TX_DRV test point · exact map; disabled output isolated; no connector-side wiring | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Exercise local buffer enable · TX_GATE/GPIO17_TX/local fixture · TX_DRV/UART_BUFFER_HIGH_Z/UART_BUFFER_FOLLOWS · capture logic traces · pin 3 high impedance disabled and follows pin 2 enabled | |

**Relay exerciser identity:** ____________________ **Firmware identity:** ____________________

**STOP gate:** TX_DRV drives while disabled, enabled waveform does not follow, any connector-side wire is installed, or exerciser identity is absent.

**PASS gate:** Local high impedance and following behavior pass; connector-side checks remain deferred. **PASS — continue** to Cluster 9 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

# Cluster 9 — Indicators and VBUS sensing

| | |
|---|---|
| **Dependencies:** Cluster 3<br>**Inputs:** LOGIC_3V3, GPIO38, USB_5V, GND | **Outputs:** POWER_LED, STATUS_LED, VBUS_PRESENT_N<br>**Source state:** USB-only test then standalone-only test; never join VBUS to a local rail |

## Parts

**Q2** 2N7000 · 4878-2N7000CT-ND; **D4** green LTL-4233 · 160-1130-ND; **D5** red 151051RS11000 · 732-5016-ND; 330 Ω, 1 kΩ, and 10 kΩ resistors · NOT ORDERED; 100 nF · BC1084CT-ND.

**Operator part record:** Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330 Ω/1 kΩ/10 kΩ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode
**Operator wiring record:** Q2/D4/D5/resistors/C7 · Q2 G/D/S, LED anode/cathode, resistor/capacitor endpoints · USB_5V/GND/VBUS_PRESENT_N/LOGIC_3V3/GPIO38/POWER_LED/STATUS_LED · VIOLET/BLACK/GREEN/BLUE/YELLOW

**Operator bypass wiring record:** C7 · leads 1/2 · LOGIC_3V3/GND · BLUE/BLACK

## Build — exact pin/net/color wiring

VIOLET DevKit 5V/VBUS to Q2 gate with 10 kΩ discharge to GND; BLACK Q2 source to GND; GREEN drain to GPIO7/VBUS_PRESENT_N with BLUE 10 kΩ pull-up to LOGIC_3V3. YELLOW GPIO38 through 330 Ω to green LED anode; BLUE LOGIC_3V3 through 1 kΩ to red LED anode; both cathodes BLACK to GND. Wire C7 lead 1 to LOGIC_3V3 and C7 lead 2 to GND as the 100 nF logic bypass.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Verify transistor leads and LED polarity · polarity · Q2 G/D/S, both LED anodes/cathodes · exact map; VBUS isolated from all local positive rails | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Toggle source and status · USB_5V/LOGIC_3V3/GPIO38 · VBUS_PRESENT_N/LEDs · record truth/voltage · VBUS sense active-low; USB present low, USB absent high; red power and green status indications correct | |

**Bound operator part record:** Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330 Ω/1 kΩ/10 kΩ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

VBUS sense is active-low and isolated: no connection may ever join VBUS to a local rail.

**STOP gate:** VBUS joins a local rail, active-low truth is wrong, LED polarity/current limiting is wrong, or sources overlap.

**PASS gate:** Both LEDs and isolated active-low VBUS sense pass under exclusive sources. **PASS — continue** to Cluster 10 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

# Cluster 10 — Whole-device standalone bench test

**Dependencies:** Cluster 1, Cluster 2, Cluster 3, Cluster 4, Cluster 5, Cluster 6, Cluster 7, Cluster 8, Cluster 9
**Inputs:** +8V_RAW, GND, observer firmware

**Outputs:** standalone boot, Wi-Fi observation, event log, UART idle evidence
**Source state:** USB physically absent; STANDALONE POWER installed; observer verified; relay off; TX disabled

**Outputs record:** standalone boot · Wi-Fi observation · event log · UART idle evidence

## Parts

**ASSEMBLY** completed Clusters 1–9 · NOT ORDERED · inspect all removable links and polarities. **Firmware artifact** observer image and manifest · NOT ORDERED · exact immutable identity required.

**Operator part record:** ASSEMBLY/FW · completed Clusters 1–9; observer image and manifest · whole device; immutable diagnostic artifact · NOT ORDERED · inspect links/polarities; observer manifest proves relay and TX disabled
**Operator wiring record:** ASSEMBLY/FW · all named inter-cluster endpoints and STANDALONE POWER terminals · +8V_RAW/VIN/TSR_3V3/LOGIC_3V3/GND · RED/ORANGE/BLUE/BLACK

## Build — exact pin/net/color wiring

With all power off, inspect every named inter-cluster wire; remove USB physically; install BLUE STANDALONE POWER from TSR_3V3 to LOGIC_3V3; leave relay command off and TX disabled. No new uncontrolled connection is permitted.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Final standalone source audit · resistance · +8V_RAW/VIN/TSR_3V3/LOGIC_3V3/+5V_RLY to GND · no shorts; USB absent; source handoff exact | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Boot observer from raw input · +8V_RAW · standalone boot/Wi-Fi/event log/UART idle · record rails, reset log, GPIO16/GPIO18 idle · VIN 7.20–7.90 V; TSR_3V3 and LOGIC_3V3 3.20–3.40 V; relay off, feedback (0,1), TX disabled | |

**Firmware identity:** ____________________ **Observer identity:** ____________________

**Observer manifest:** ____________________ (path or SHA proving relay and TX outputs disabled)

## Dedicated UART idle evidence

| Evidence field | Measured | State |
|---|---|---|
| GPIO16 UART idle (V) | | Before UART configuration |
| GPIO18 UART idle (V) | | Before UART configuration |

**STOP gate:** Identity/manifest missing, rail out of range, reset, Wi-Fi/event-log failure, UART idle missing, relay enables, TX enables, or feedback differs from (0,1).

**PASS gate:** Identified observer boots standalone and all passive observations pass. **PASS — continue** to Cluster 11 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

# Cluster 11 — RJ45 pass-through and treadmill bypass

**Cluster 11 is the end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass proof.**

| | |
|---|---|
| **Dependencies:** Cluster 7, Cluster 8, Cluster 10<br>**Inputs:** CONSOLE.1–8, MOTOR.1–8, verified observer state | **Outputs:** eight mapped conductors, end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass, bypass-only treadmill evidence<br>**Source state:** USB absent; standalone installed; observer verified; relay off; TX disabled; source → state → harness → direct-path sequence |

## Parts

**J1/J2** CONSOLE/MOTOR RJ45 breakouts · NOT ORDERED · meter-map terminal numbers. **Harness** reviewed dual-pin fused-DMM current harness · NOT ORDERED. **K1** verified Cluster 7 relay contacts · 39-G5V-2DC5-ND.

**Operator part record:** J1/J2/HARNESS/K1 · CONSOLE/MOTOR RJ45 breakouts; dual-pin fused-DMM harness; verified relay · two 8P8C breakouts; reviewed current harness; G5V-2 · NOT ORDERED; 39-G5V-2DC5-ND · meter-map connector terminals; verify fused jack/polarity; relay bottom-view map proven
**Operator wiring record:** J1/J2/K1 · RJ45 pins 1–8; K1 pins 4/6/8 · GND/+8V_RAW/PIN3/PIN4_PASS/PIN5_SAFETY/CONSOLE.6/MOTOR.6/TX_DRV · BLACK/RED/WHITE/YELLOW

## Build — exact pin/net/color wiring

1. **Isolated mapping first:** Keep all J1/J2 terminals unwired and all common links open; map each conductor independently, CONSOLE.1–8 and MOTOR.1–8, and verify unrelated pins open before any common connection. Do not continue until the independent map is signed.
2. **Only after mapping passes:** Only after the isolated map is signed, join pins 1/7 and 2/8; then wire direct paths, K1 contacts, TX_DRV, GPIO18_RX, and GPIO16_RX by named net and color. Use BLACK J1/J2 pins 1 and 7 to GND; RED pins 2 and 8 to +8V_RAW; WHITE pin 3↔3, 4↔4, 5↔5; YELLOW CONSOLE.6 to K1 pin 4, MOTOR.6 to K1 pin 6, K1 pin 8 to TX_DRV; GREEN CONSOLE.6 through 10 kΩ to GPIO18_RX and PIN3 through 10 kΩ to GPIO16_RX.

## Unpowered test

| Instruction / kind / points / expected | Evidence |
|---|---|
| Map every connector conductor independently · mapping · CONSOLE.1–8 to MOTOR.1–8 and K1 pins 4/6/8 · each intended path below 2 Ω, unrelated paths open; end-to-end CONSOLE.6 ↔ MOTOR.6 through NC | |

## Powered test

| Instruction / input / output / evidence / limits | Measured |
|---|---|
| Run bypass-only treadmill sequence · treadmill +8 V through fused harness · current/drop/receive-only UART/thermal · record source state, harness, restored direct paths, endpoints · separately measure treadmill current no more than 500 mA; supply drop no more than 50 mV; ground-return drop no more than 50 mV; 15-minute endpoints no more than 40°C and no more than 10°C over ambient | |

**Firmware identity:** ____________________ **Observer identity:** ____________________

**Observer manifest:** ____________________ (path or SHA proving relay and TX outputs disabled)

## Bypass-only controlled sequence

1. All sources off before RJ45 changes. Attach cables only after Clusters 1–10 signed PASS.
2. Verify bypass-only source state: USB absent; STANDALONE POWER installed; observer verified; relay off; TX disabled.
3. With all sources off before harness changes, remove both independent direct +8V paths and install the reviewed dual-pin fused-DMM harness. Measure current separately at no more than 500 mA.
4. Remove power before removal of the harness. Verify zero current, remove harness, then restore both independent +8V paths and continuity-check each.
5. Voltage drop is measured only after direct-path restoration. Thermal test only after direct-path restoration; then run the fifteen-minute hold.

## Cluster 11 — dedicated bypass and thermal evidence

| Evidence field | Measured | Limit/state |
|---|---|---|
| Treadmill current (mA) | | ≤500 mA, fused harness |
| Supply drop (mV) | | ≤50 mV after direct paths restored |
| Ground-return drop (mV) | | ≤50 mV after direct paths restored |

| RJ45 pin | CONSOLE temperature (°C) | MOTOR temperature (°C) | Limit after 15 minutes |
|---|---|---|---|
| 1 — GND | | | ≤40°C; ≤10°C rise |
| 2 — +8 V | | | ≤40°C; ≤10°C rise |
| 3 | | | ≤40°C; ≤10°C rise |
| 4 | | | ≤40°C; ≤10°C rise |
| 5 | | | ≤40°C; ≤10°C rise |
| 6 — switched data | | | ≤40°C; ≤10°C rise |
| 7 — GND | | | ≤40°C; ≤10°C rise |
| 8 — +8 V | | | ≤40°C; ≤10°C rise |

| Relevant breadboard power endpoint | Measured | Limit |
|---|---|---|
| +8V endpoint temperature (°C): hottest exact endpoint ___________ | | ≤40°C; ≤10°C rise |
| GND endpoint temperature (°C): hottest exact endpoint ___________ | | ≤40°C; ≤10°C rise |

The current no-control diagnostic cannot pass the relay exercise gate and cannot pass the UART exercise gate. This is observation only. No treadmill transfer and no MCU TX. A future procedure must require qualified functional-firmware identity and matching production safety evidence; bypass PASS grants no transfer permission.

**STOP gate:** Any map, identity, current, drop, receive-only UART, bypass, thermal, or reset check fails. Power off; do not transfer the relay or transmit.

**PASS gate:** All eight paths, NC bypass, current, restored-path drop, receive-only observation, and thermal evidence pass. **PASS — continue** only to documentation closeout, never to treadmill control.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________